RIZ MUGHAL



QUIZ MASTER

MTH501(28 TO 35)

100% correct solution.

For more information you can visit my channel and for any type of help related to CS619 you can contact me.

YOUTUBE CHANNEL:

https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists

MTH501:Quiz#2 Quiz Start Time: 10:18 AM, 03 Se

Question # 1 of 10 (Start time: 10:18:39 AM, 03 September 2020)

A system of linear equations is said to be homogeneous if it can be written in the form _____.

Riz Mughal

Select the correct option

Reload Mat

- ○ AX=B
- ◉ AX=0

Click to Save Answer & Move to Nex

MTH501:Quiz#2 Quiz Start Time: 10:18 AM, 03 Septeml

Question # 2 of 10 (Start time: 10:19:03 AM, 03 September 2020) Total

If $AB = I = BA$ for $matrices A, B \; and \; I, where \; I \; is \; an \; identity \; matrix, \; then$

Select the correct option

Reload Math Equ

○	B is inverse of A.
○	A is inverse of B.
○	$A^{(-1)} = B, B^{(-1)} = A,$
◉	All of the above.

Riz Mughal

MTH501:Quiz#2

Question # 3 of 10 (Start time: 10:19:22 AM, 03 September 2020)

A square matrix A is said to be diagonal if A is similar to a matrix

Select the correct option

Riz Mughal

○	Column matrix
○	Zero matrix
◉	Diagonal matrix
○	None of the above

MTH501:Quiz#2 Quiz Start Time: 10:18 AM, 03 September 2

Question # 4 of 10 (Start time: 10:19:40 AM, 03 September 2020) Total Mark

Let A be the matrix of order 2X3 and B be the matrix of order 3X5,then which of the following is the order of the matrix AB?

Select the correct option

Reload Math Equatio

- 2X3
- 3X5
- 3X3
- 2x5

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 10:18 AM, 03 September 2020

Question # 5 of 10 (Start time: 10:20:04 AM, 03 September 2020) **Total Marks:**

Let 'Ax = 0' be a homogeneous linear system of 'n' equations and 'n' unknowns. Then, the coefficient matrix 'A' is invertible if and only if this system has _____ solution.

Riz Mughal

Select the correct option

Reload Math Equations

- ○ No
- ◉ trivial
- ○ non-trivial
- ○ infinite many

MTH501:Quiz#2 Quiz St

Question # 6 of 10 (Start time: 10:20:25 AM, 03 September 2020)

If x - 2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is

Select the correct option

- (●) 2
- ( ) -2
- ( ) 1/2
- ( ) 0

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 10:18 AM, 03 Septemb

Question # 7 of 10 (Start time: 10:20:42 AM, 03 September 2020) Total M

If $\lambda + 2$ is a factor of the characteristic polynomial of matrix C , then which of the following is the Eigenvalue of C ?

Select the correct option

Reload Math Equa

- 2
- -2
- $\frac{1}{2}$
- 0

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 10:18 AM, 03 September 2020

Question # 8 of 10 (Start time: 10:21:02 AM, 03 September 2020) **Total Marks:** 1

Let A and B be the square matrices. Then, A and B are invertible with $B = A^{-1}$ and $A = B^{-1}$ if and only if $AB = BA$ equals to a (an) _____ matrix.

Select the correct option

Reload Math Equations

- ○ Singular
- ○ Square
- ◉ Identity
- ○ Rectangular

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 10

Question # 9 of 10 (Start time: 10:21:20 AM, 03 September 2020)

Let V be a five-dimensional vector space, and let S be a subset of V which spans V. Then S

Select the correct option

Riz Mughal

○	Must be linearly dependent
○	Must be a basis for V
○	Must have infinitely many elements
◉	Must have at most five elements

MTH501:Quiz#2

Question # 10 of 10 (Start time: 10:21:38 AM, 03 September 2020)

If u + v =u + w, then:

Select the correct option

- v + w.
- v ≠ w
- v = w
- None of the above.

Riz Mughal

2nd account

MTH501:Quiz#2 Qu

Question # 1 of 10 (Start time: 10:25:52 AM, 03 September 2020)

If one of the eigenvalues of $[A]_{n\times n}$ is zero, it implies--------

Riz Mughal

Select the correct option

○	The solution to $[A][X] = [C]$ a system of equations is unique
●	The determinant of $[A]$ is zero
○	The solution to $[A][X] = [0]$ system of equations is trivial
○	The determinant of $[A]$ is nonzero

MTH501:Quiz#2 Quiz Start Time: 10:25 AM, 03 Septe

Question # 2 of 10 (Start time: 10:26:19 AM, 03 September 2020) Tot

Le a matrix A has both negative and positive eigen values, so in this case origin behaves as a ---------- point

Riz Mughal

Select the correct option

Reload Math E

- (●) Saddle
- ( ) Critical

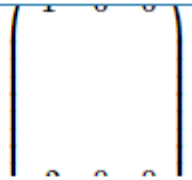
MTH501:Quiz#2

Question # 3 of 10 (Start time: 10:26:46 AM, 03 September 2020)

Which one of the following is a null matrix?

Riz Mughal

Select the correct option

○ 

◉ $\begin{pmatrix} 0 & 0 & 0 \end{pmatrix}$

MTH501:Quiz#2 Quiz Start Ti

Question # 4 of 10 (Start time: 10:27:16 AM, 03 September 2020)

If x+2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is

Select the correct option

○	2
◉	-2
○	1/2
○	0

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 10:25 AM, 03 September 2020

Question # 5 of 10 (Start time: 10:27:35 AM, 03 September 2020) **Total Marks:** 1

If λ is an eigenvector of A, then every nonzero vector x such that $Ax = \lambda x$ is called an ---------- of A corresponding to ----------

Select the correct option

Reload Math Equations

Riz Mughal

- ○ Eigenvalue, λ
- ◉ Eigenvector, λ
- ○ Eigenvalue, A
- ○ Eigenvector, A

MTH501:Quiz#2

Question # 6 of 10 (Start time: 10:27:55 AM, 03 September 2020)

Let A be n x n matrix, then A is invertible if and only if

Select the correct option

- (●) det A is not zero
- ( ) det A is zero

Riz Mughal

MTH501:Quiz#2

Question # 7 of 10 (Start time: 10:28:13 AM, 03 September 2020)

The Invertible Matrix Theorem applies only to _____ matrices.

Select the correct option

- ○ Rectangular
- ◉ Square
- ○ Identity
- ○ Scalar

Riz Mughal

MTH501:Quiz#2

Question # 8 of 10 (Start time: 10:28:34 AM, 03 September 2020)

A 3 x 3 identity matrix have three and _____eigenvalues.

Riz Mughal

Select the correct option

- (●) same
- ( ) distinct

MTH501:Quiz#2 Quiz Start Time: 10:25 AM, 03

Question # 9 of 10 (Start time: 10:28:53 AM, 03 September 2020)

An $n \times n$ matrix A is said to be diagonalizable if and only if A has n ---------- eigenvectors.

Riz Mughal

Select the correct option

Reload

- ○ Linearly dependent
- ◉ Linearly Independent

MTH501:Quiz#2

Question # 10 of 10 (Start time: 10:29:13 AM, 03 September 2020)

What is Eigen value?

Select the correct option

- ○ A vector obtained from the coordinates
- ○ A matrix determined from the algebraic equations
- ◉ A scalar associated with a given linear transformation
- ○ It is the inverse of the transform

Riz Mughal

3rd account

MTH501:Quiz#2

Question # 1 of 10 (Start time: 12:58:09 PM, 03 September 2020)

A column replacement operation on A does not change the _____.

Select the correct option

- (●) determinant
- ( ) matrix
- ( ) row
- ( ) column

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 12:58 PM, 03

Question # 2 of 10 (Start time: 12:58:37 PM, 03 September 2020)

Le a matrix A has both negative and positive eigen values, so in this case origin behaves as a ---------- point

Select the correct option

Reload M

- (●) Saddle
- ( ) Critical

Riz Mughal

MTH501:Quiz#2

Question # 3 of 10 (Start time: 12:58:59 PM, 03 September 2020)

A null space is a vector space.

Select the correct option

- (●) True
- ( ) False

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 12:58 PM, 03 Septembe

Question # 4 of 10 (Start time: 12:59:27 PM, 03 September 2020) Total M

Each pair of eigenvalue and its corresponding eigenvector provides a solution of the equation $x' = Ax$ which is called ---- ------ of the differential equation.

Select the correct option

Reload Math Equa

- Eigensolution
- Eigenfunction

Riz Mughal

MTH501:Quiz#2 Quiz

Question # 5 of 10 (Start time: 12:59:50 PM, 03 September 2020)

If x - 2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is

Select the correct option

- (●) 2
- ( ) -2
- ( ) 1/2
- ( ) 0

Riz Mughal

MTH501:Quiz#2

Question # 6 of 10 (Start time: 01:00:07 PM, 03 September 2020)

A row replacement operation on A does not change the _____.

Select the correct option

- (●) determinant
- ( ) matrix
- ( ) columns
- ( ) rows

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 12:58 PM.

Question # 7 of 10 (Start time: 01:00:26 PM, 03 September 2020)

If A is invertible and b in R^n be any vector. Then, we must have a matrix $A^{-1}b$,which is a solution of _____

Riz Mughal

Select the correct option

Reload

○	$A^{-1}x = b$
○	$A^2x = b$
○	$A^tx = b$
◉	$Ax = b$

MTH501:Quiz#2 Quiz St

Question # 8 of 10 (Start time: 01:01:04 PM, 03 September 2020)

What is the maximum possible number of pivots in a 6x6 matrix?

Select the correct option

- ○ 0
- ○ 2
- ○ 4
- ◉ 6

Riz Mughal

MTH501:Quiz#2

Question # 9 of 10 (Start time: 01:01:54 PM, 03 September 2020)

If one of the Eigenvalues of $[A]_{n\times n}$ is zero, it implies

Select the correct option

○	The solution to [A][X]=[C] system of equations is unique
◉	The determinant of [A] is zero
○	The solution to [A][X]=[0]
○	The determinant of [A] is nonzero

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 12:58

Question # 10 of 10 (Start time: 01:02:15 PM, 03 September 2020)

If 3 is an eigenvalue of A and x is a corresponding eigenvector, then what is the eigenvalue of A^2 ?

Riz Mughal

Select the correct option

○	12
◉	9
○	6
○	3

4th account

MTH501:Quiz#2

Question # 1 of 10 (Start time: 01:10:21 PM, 03 September 2020)

What is the maximum possible number of pivots in a 6x6 matrix?

Select the correct option

○	0
○	2
○	4
●	6

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 01:10 PM, 03 Septembe

Question # 2 of 10 (Start time: 01:10:41 PM, 03 September 2020) Total M

The characteristics polynomial of a 3x 3 identity matrix is _____, if x is the eigen values of the given 3 x 3 identity matrix.

Select the correct option

- (x - 1)^3
- (x + 1)^3
- X^3
- (1 - x)^3

Riz Mughal

MTH501:Quiz#2 Quiz

Question # 3 of 10 (Start time: 01:11:01 PM, 03 September 2020)

Which one of the following is a diagonal matrix?

Select the correct option

Riz Mughal

- $\begin{pmatrix} 2 & 0 & 0 \end{pmatrix}$
- $\begin{pmatrix} 2 & 2 & 0 \end{pmatrix}$

MTH501:Quiz#2 Quiz Start Time: 01:10 PM, 03 September 2020

Question # 4 of 10 (Start time: 01:11:21 PM, 03 September 2020) **Total Marks:** 1

A partitioned matrix 'A' is said to be block diagonal if the matrices on the main diagonal are square and all other position matrices are _____.

Select the correct option

- (●) Zero
- ( ) Unit
- ( ) Nonzero symmetric
- ( ) Nonzero skew symmetric

Riz Mughal

MTH501:Quiz#2 Qu

Question # 5 of 10 (Start time: 01:11:59 PM, 03 September 2020)

If A is an invertible square matrix then

Select the correct option

- [x] $(A^T)^{-1} = (A^{-1})^T$
- [ ] $(A^T)^T = (A^{-1})^T$
- [ ] $(A^T)^{-1} = (A^{-1})^{-1}$
- [ ] None of the above

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 01:10 PM, 03 September 2020

Question # 6 of 10 (Start time: 01:12:22 PM, 03 September 2020) **Total Marks:** 1

A blocked matrix in which blocks (blocked matrices) are repeated down the diagonals of the matrix is called a _____ matrix.

Riz Mughal

Select the correct option

Reload Math Equations

- ○ Blocked square
- ◉ Blocked diagonal-constant
- ○ Blocked identity
- ○ Blocked rectangular

MTH501:Quiz#2

Question # 7 of 10 (Start time: 01:12:42 PM, 03 September 2020)

Two equivalent vectors must have the same initial point:

Select the correct option

- ○ true
- ◉ false
- ○ May be
- ○ None of the above

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 01:10 PM, 03 September

Question # 8 of 10 (Start time: 01:13:04 PM, 03 September 2020) Total Mar

If A= transpose of A(where A is a square matrix), then which of the following is the most appropriate option for A?

Select the correct option

Reload Math Equati

- A is invertible matrix.
- A is singular matrix.
- A is symetric matrix.
- A is a scalar matrix.

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 01:10 PM, 03 Septe

Question # 9 of 10 (Start time: 01:13:24 PM, 03 September 2020) Tota

\lambda is an eigenvalue of a matrix A if and only if the equation $(A - \lambdaI)x = 0$ has a ---------

Select the correct option

Reload Math E

- (●) Non-trivial solution
- ( ) Trivial Solution

Riz Mughal

MTH501:Quiz#2 Quiz Star

Question # 10 of 10 (Start time: 01:13:49 PM, 03 September 2020)

Which of the following is (are) the Eignvalue(s) of the matrix?

$$A = \begin{bmatrix} 3 & 0 \\ \backslash 5 & 3 \end{bmatrix}$$

Select the correct option

- (●) 3
- ( ) 5
- ( ) 3, 4
- ( ) 3, 5

Riz Mughal

5th account

MTH501:Quiz#2 Quiz Start Time: 03:41 PM, 03 September 2

Question # 1 of 10 (Start time: 03:41:40 PM, 03 September 2020) Total Mark

The complex conjugate of a complex vector$x$ in C^n is the vector $\bar{x}$ in C^n whose entries are the ---------- conjugates of the entries in x.

Select the correct option

Reload Math Equatio

- Real
- Complex

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 03:41 PM, 03 Sept

Question # 2 of 10 (Start time: 03:42:11 PM, 03 September 2020) To

If A= transpose of A(where A is a square matrix), then which of the following is the most appropriate option for A?

Select the correct option

Reload Math

- A is invertible matrix.
- A is singular matrix.
- A is symetric matrix.
- A is a scalar matrix.

Riz Mughal

MTH501:Quiz#2

Question # 3 of 10 (Start time: 03:42:30 PM, 03 September 2020)

Multiplication of a partitioned matrix by a scalar is also computed _____.

Select the correct option

○	Row by row
○	Column by column
○	Diagonal by diagonal
◉	Block by block

Riz Mughal

MTH501:Quiz#2 Quiz Start Tir

Question # 4 of 10 (Start time: 03:42:47 PM, 03 September 2020)

If the real part of the eigenvalue is zero, then the trajectories form ---------- around the origin.

Select the correct option

- ○ Parabola
- ○ Hyperbola
- ◉ Ellipse
- ○ None of these

Riz Mughal

MTH501:Quiz#2

Question # 5 of 10 (Start time: 03:43:04 PM, 03 September 2020)

A row interchange _____ the sign of the determinant.

Riz Mughal

Select the correct option

- (●) changes
- ( ) does not change

MTH501:Quiz#2 Quiz Start Time: 03:41 PM

Question # 6 of 10 (Start time: 03:43:20 PM, 03 September 2020)

A system of linear equations is said to be homogeneous if it can be written in the form _____.

Select the correct option

Reloa

- AX=B
- AX=0

Riz Mughal

MTH501:Quiz#2

Question # 7 of 10 (Start time: 03:43:36 PM, 03 September 2020)

If u + v =u + w, then:

Select the correct option

- v + w.
- v ≠ w
- v = w
- None of the above.

Riz Mughal

MTH501:Quiz#2

Question # 8 of 10 (Start time: 03:43:52 PM, 03 September 2020)

A 3 x 3 identity matrix have three and _____eigen values.

Select the correct option

- (●) same
- ( ) distinct

Riz Mughal

MTH501:Quiz#2

Question # 9 of 10 (Start time: 03:44:09 PM, 03 September 2020)

A column replacement operation on A does not change the _____

Select the correct option

- (●) determinant
- ( ) matrix
- ( ) row
- ( ) column

Riz Mughal

MTH501:Quiz#2 Quiz Start Time: 03:41 PM, 03 September 202

Question # 10 of 10 (Start time: 03:44:25 PM, 03 September 2020) **Total Marks:**

A blocked matrix in which blocks (blocked matrices) are repeated down the diagonals of the matrix is called a _____ matrix.

Select the correct option

Reload Math Equations

- Blocked square
- Blocked diagonal-constant
- Blocked identity
- Blocked rectangular

Riz Mughal